

Slide 1

Why Inheritance Fails (UML-1 Reference)

- Naive inheritance leads to rigid, hard-to-maintain designs.
 - **SRP Broken:** One class handles video, quality, network, battery → too complex.
 - **OCP Broken:** Adding features requires modifying existing classes → unstable.
 - **LSP Broken:** Subclasses may not fully support parent behavior → unexpected bugs.
 - **DIP Broken:** High-level modules depend on concrete classes → hard to test & scale.
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Slide 2

Class Explosion & Developer Pain

- Multiple dimensions (Device, Quality, Network, Battery, AI) → 50–200 classes.
- Adding new behavior requires modifying many classes → high regression risk.
- Code duplication across subclasses → harder to fix bugs.
- Inheritance locks behavior at runtime → cannot adapt mid-stream.
- Developer issues: slow feature rollout, complex debugging, testing nightmares.